EESC4550/5550: Practical Atmospheric Modeling (Fall 2026)
Homework 5 – Decision Trees, SHAP, Deweathering
Total Score: 60

Learning Objectives:

- Build and tune tree-ensemble regressors (Random Forest, Gradient Boosting).
- Compute SHAP values for tree models and interpret them globally and locally.
- Use SHAP dependence plots to identify nonlinear and threshold effects.
- Compute SHAP interaction values and identify physically meaningful feature pairs.
- Build a meteorology-only model that predicts daily mean O_3 without using any air-quality predictors or time indices, and construct a deweathered O_3 series by subtracting the meteorological SHAP contributions from the observations.

Part 0. Dataset Specifications

Again, we will use `hw4_data.zip` for this homework assignment. Inside the `.zip` file, there are **10** `.csv` data files. These files are also available at `/lustre/project/eesc4550/shared/hw4`.

Data source: air quality data is downloaded from https://cd.epic.epd.gov.hk/EPICDI/air/station/?lang=en, and the meteorological data is downloaded from https://data.gov.hk/en-datasets/category/climate-and-weather.

Target: Ozone concentration at Tap Mun, recorded in `air_daily.csv`

File	Contents	Freq.	Units
`air_daily.csv`	Hourly air-quality observations (Tap Mun)	Daily	$\mu g/m^3$ (CO in 10 $\mu g/m^3$)

Meteorological Predictors:

File	Variable	Freq.	Units
`daily_HKO_CLD_ALL.csv`	Daily mean cloud amount	Daily	%
`daily_HKO_CLMMAXT_ALL.csv`	Daily maximum temperature	Daily	°C
`daily_HKO_DEW_ALL.csv`	Daily mean dew point temperature	Daily	°C
`daily_HKO_MSLP_ALL.csv`	Daily mean sea-level pressure	Daily	hPa
`daily_HKO_RF_ALL.csv`	Daily total rainfall	Daily	mm
`daily_HKO_RH_ALL.csv`	Daily mean relative humidity	Daily	%
`daily_KP_UV_ALL.csv`	Mean UV index (7 a.m.-6 p.m.) at King's Park	Daily	index
`daily_TME_PDIR_ALL.csv`	Prevailing wind direction at Tap Mun East	Daily	degrees
`daily_TME_WSPD_ALL.csv`	Mean wind speed at Tap Mun East	Daily	km/h

Air Pollutant Predictors: O_3, CO and NOx concentration at Tap Mun in `air_daily.csv`.

Note: O_3 plays two roles in this assignment. It is the **target** (daily mean O_3 on day t) and it is also a **predictor** (daily mean O_3 on day $t-1$ as an input to the model) since ozone is **highly persistent** from one day to the next.

For this homework, we consider the period from 2017/09/01 to 2026/08/31. Use the same dataset as what you have done for HW4.

Part 1. Random Forest

Question 1 (6 pts). Fit a `RandomForestRegressor` on the training set. Tune at least three of the following hyperparameters using the validation set: `n_estimators`, `max_depth`, `min_samples_leaf`, `max_features`. Justify your selection principles and select the best configuration by validation RMSE. Report on the test set with RMSE, MAE, and R^2. Produce a **time series** of predicted vs observed daily mean O_3 on the dates associated with the test set.

Part 2. Gradient Boosting

Question 2 (6 pts). Using the XGBoost model, fit a gradient-boosted tree model on the training set. Tune at least three of the following hyperparameters using the validation set: `n_estimators`, `max_depth`, `learning_rate`, `max_depth`, `reg_lambda`. Also, use `early_stopping_rounds` with the validation set to stop boosting when validation RMSE stops improving. Justify your selection principles and select the best configuration by validation RMSE.
Report on the test set with RMSE, MAE, and R^2. Produce a **time series** of predicted vs observed daily mean O_3 on the dates associated with the test set.

Part 3. SHAP Interpretations

Question 3 (15 pts). Use the best tree model from Question 2. Compute SHAP values on the test set using `shap.TreeExplainer`. Produce a SHAP summary plot (beeswarm) and SHAP bar plot (mean absolute SHAP value per feature). Which features dominate globally? What are the directions of their effects? Do the top-3 features and their effects match atmospheric chemistry and meteorological expectations? Do any features show threshold or non-monotonic behaviour? (e.g. NO_x / O_3 titration)

Question 4 (13 pts). Select two days from the test set, which corresponds to the day with the highest observed O_3 and the day with the lowest observed O_3, respectively. For each day, produce a SHAP force plot (or waterfall plot). For each day, write a short paragraph explaining: which features pushed the prediction up or down and by how much ($\mu g/m^3$), and whether the explanation matches the meteorology and chemistry on that day. Also, compare the two days and describe the key differences in feature contributions?

Part 4. Deweathering Model

The deweathering model must use meteorology only. Specifically, you are allowed to use the meteorological predictors `TMAX`, `CLD`, `DEW`, `MSLP`, `RF`, `RH`, `UV`, `WSPD`, `WIND_SIN`, `WIND_COS` (from `WDIR`), and time predictors `DOY_SIN`, `DOY_COS` (day-of-year harmonics). However, you are not allowed to use any of the O_3, NO_x, CO data (any lag of any of them), year, month, or any monotonic time index, or any features derived from the target as predictors.

Do not change the dataset, station, or study period, the target definition, and the chronological 70 / 15 / 15 split.

Question 5 (20 pts). Again, fit an XGBoost model on the revised training set and report validation and test metrics for the model with RMSE, MAE, R^2 $(\mu g/m^3)$. Compute SHAP values on the full dataset (train + validation + test) using `shap.TreeExplainer`. Let ϕ_0 be the expected value (`explainer.expected_value`) and $\phi_j(yr, t)$ be the SHAP contribution of feature j on day t of the year yr.

The SHAP decomposition can be written as:

$$\hat{y}(yr, t) \approx \phi_0 + \sum_{j \in \text{time}} \phi_j\,(yr, t) + \sum_{j \in \text{met}} \phi_j\,(yr, t)$$

Finally, compute the deweathered O_3, where the individual pollution episodes and day-to-day fluctuations are smoothed out, but it is still allowed to capture sub-seasonal variations in the background:

$$y_{\text{deweathered}}(yr, t) = \phi_0 + \sum_{j \in \text{time}} \phi_j\,(yr, t) + \frac{1}{11}\sum_{p=0}^{9}\frac{1}{15}\sum_{d=-7}^{+7}\sum_{j \in \text{met}} \phi_j(p + 2017, t + d)$$

Note that if there are any missing values for meteorological predictors so that the SHAP values cannot be computed, we will instead take average over the available SHAP values for deweathering. You may also interpolate the meteorological values between two days with available data to avoid any ambiguity. Produce a time-series plot of observed and deweathered O_3 from 2017-09-01 to 2026-08-31 and a monthly box plot of the deweathered series. Also, report the correlation between observed and deweathered series. Is the deweathered O_3 series rising, falling, or flat over 2017–2026? Discuss at least three mechanisms that could explain the deweathered trend.

Submission Guideline: To complete the homework, please submit **TWO** files:

- **`hw5_<your_id>.pdf`**, which contains the answers to all the **6** questions;
- **`hw5_<your_id>.zip`**, which contains the Python scripts to produce the included plots and scripts that perform data analysis.

If any AI tool was used in writing or code generation, please include a short note on AI use.

----- **End of Homework 5** -----